

SMART CLASSROOM

ENERGY GUARDIAN

Automatic Energy Management System for a Classroom

Controller	Arduino UNO
Display	16×2 LCD with I2C interface
Motor/Fan Model	DC motor with L298N motor driver
Prototype	Breadboard, jumper wires, sensor/indicator modules and 9V battery

Concept: When the classroom is detected as empty, the system automatically switches OFF the connected light/fan loads, reducing unnecessary energy consumption. The LCD provides a simple status indication.

2025–2026

1. Aim

To design and build a **Smart Classroom Energy Guardian** that monitors classroom occupancy and automatically controls electrical loads such as lights and fans. The system is intended to switch OFF unnecessary loads when the classroom is empty and thereby reduce energy wastage.

2. Problem Statement

In many classrooms, lights and fans may remain ON even when students have left the room. This causes unnecessary electricity consumption and increases operating cost. Manual switching also depends on the user remembering to turn the loads OFF.

The proposed project addresses this problem by using a sensor-based controller. The controller checks whether the classroom is occupied and changes the output state automatically.

3. Proposed Solution

The proposed system uses an **Arduino UNO** as the main controller. A sensor/indicator input is used to represent classroom occupancy. Based on the detected condition, the Arduino controls the output load. In the prototype, an **L298N motor driver and DC motor** are used to demonstrate fan/load control, while a **16×2 LCD** displays the system status.

When the classroom is occupied, the system can keep the required load active. When the classroom becomes empty, the controller sends a control signal to turn the connected load OFF automatically.

4. Components Required

No.	Component	Purpose
1	Arduino UNO	Main controller
2	16×2 LCD with I2C	Displays classroom/system status
3	L298N Motor Driver Module	Drives the DC motor/load
4	DC Motor	Used as a small fan/load model
5	Breadboard	Temporary circuit assembly
6	Jumper Wires	Electrical connections
7	9V Battery / DC Supply	Prototype power source
8	Sensor / Indicator Module	Provides the condition used for occupancy detection

The component list above is based on the hardware visible in the submitted prototype photographs. The exact sensor module is kept generic because its label is not clearly readable in the images.

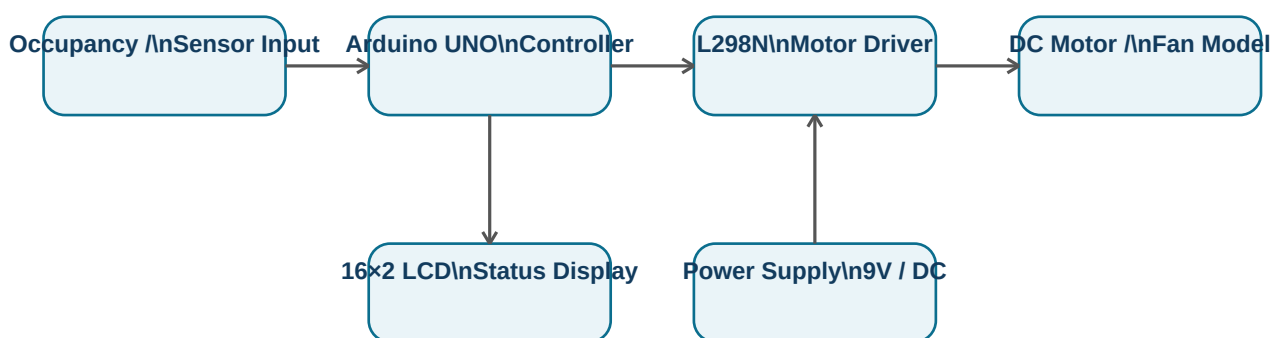
5. Circuit Working

- **Input detection:** The sensor/indicator provides a signal representing whether the classroom is occupied or empty.
- **Decision:** The Arduino UNO reads the input and compares it with the programmed condition.
- **Load control:** The Arduino sends control signals to the L298N motor driver.
- **Fan/load model:** The L298N drives the small DC motor used in the prototype as a fan/load representation.
- **Display:** The 16×2 LCD shows a simple status such as classroom occupied, load ON, classroom empty, or load OFF.
- **Energy saving:** When the classroom is empty, the controller turns OFF the connected load so that energy is not wasted.

Basic Control Logic

Classroom Condition	Controller Action	Output
Occupied	Arduino enables the required load	Light/Fan model ON
Empty	Arduino disables the load	Light/Fan model OFF

System Block Diagram



Conceptual block diagram based on the submitted prototype

6. Procedure

- Place the Arduino UNO, breadboard, LCD, L298N driver and other modules on the work table.
- Connect the required sensor/indicator input to the Arduino digital input according to the programmed pin configuration.
- Connect the 16×2 LCD through the I2C interface and provide the required power and ground connections.
- Connect the L298N motor driver to the Arduino control pins and connect the DC motor to the driver output.
- Connect the prototype power supply and make sure all grounds are common.
- Upload the Arduino program and verify that the LCD initializes correctly.
- Test the occupied condition and observe the motor/fan model response.
- Test the empty condition and verify that the connected load is switched OFF.
- Check the LCD message and repeat the test several times to confirm consistent operation.

7. Applications

- Smart classrooms – Automatically reduce unnecessary light and fan operation.
- College laboratories – Control selected electrical loads based on occupancy.
- Libraries and seminar halls – Reduce energy wastage when spaces are unused.
- Office rooms – Automate simple occupancy-based load control.
- Energy-saving demonstrations – Useful for academic and IoT/electronics prototypes.

8. Advantages

- Automatic control reduces dependence on manual switching.
- Helps reduce unnecessary electricity consumption.
- Simple Arduino-based prototype and easy to modify.
- LCD gives a clear indication of system status.
- Low-cost components can be used for a demonstration model.
- The system can be expanded with more sensors and loads.

9. Result

The Smart Classroom Energy Guardian prototype was successfully assembled using the components shown in the submitted photographs. The Arduino-based system was used to process the input condition, display the status on the LCD, and control the DC motor through the L298N driver. The prototype demonstrates the basic concept of automatically switching classroom loads according to occupancy.

10. Conclusion

The project demonstrates how a simple embedded controller can be used for classroom energy management. By detecting an empty classroom and switching OFF unnecessary loads, the system can

help avoid energy wastage. The prototype also provides practical understanding of Arduino programming, sensor input, motor-driver control, LCD interfacing, and automatic decision-making.

11. Project Demonstration

Project Video / Demo:

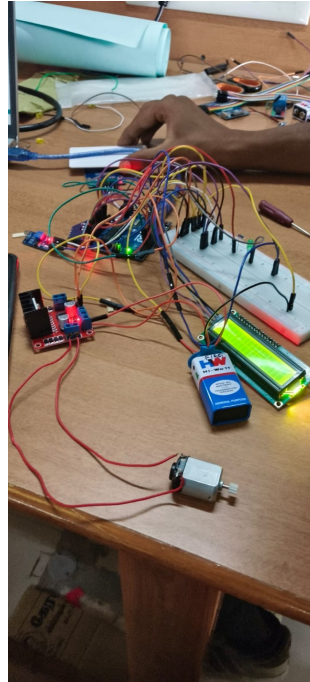
[Add Video Link Here]

12. Circuit Diagram

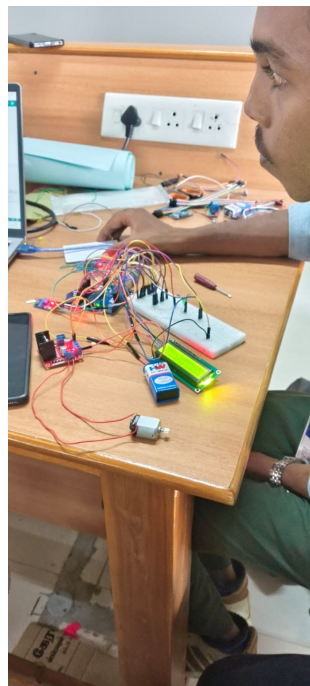
The exact pin-to-pin circuit diagram can be added after confirming the final sensor type and Arduino pin numbers. The block diagram on Page 4 represents the functional connection of the prototype.

[Add Final Circuit Diagram Here]

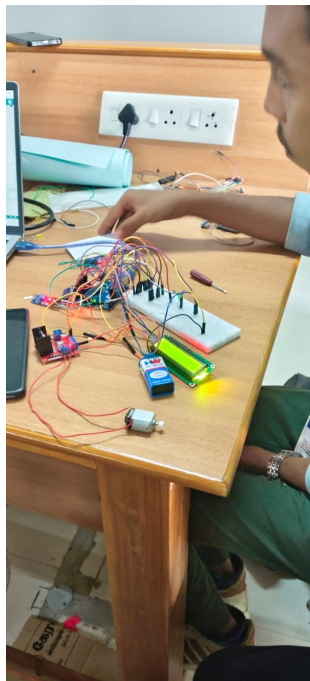
13. Project Images



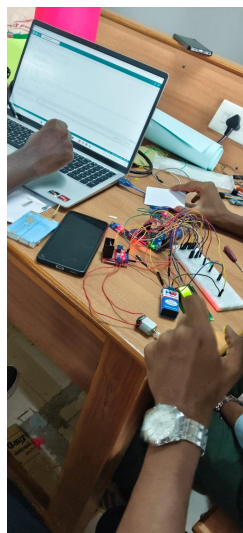
Circuit setup showing Arduino UNO, breadboard, L298N driver, LCD, battery and DC motor.



Prototype during testing.



Working prototype with LCD and motor/load section.



Team working on the Smart Classroom Energy Guardian prototype.



Team demonstration and project setup.